

BANK IT EXAM

QUESTION PATTERN & TOPIC ANALYSIS

Combined Bank & Bangladesh Bank Exams | 2020 – 2026

Total Questions Analyzed: 415 | Exams Covered: 16 | Years: 7

1. EXECUTIVE SUMMARY

This report presents a deep analysis of **415** questions from **16 written examinations** conducted by Combined Banks and Bangladesh Bank between **2020 and 2026**. The questions were collected from exams held for Officer IT, Senior Officer IT, Assistant Programmer, Assistant Network Engineer, Assistant Hardware Engineer, Assistant Database Admin, and related technical posts.

KEY STATISTICS

415 total questions analyzed from 2020–2026
16 separate exam sets across 7 years
14 distinct technical topic categories
Top 4 topics account for 68% of all questions
Networking alone = 22.9% of all questions

2. OVERALL QUESTION DISTRIBUTION BY CATEGORY

The table below ranks all 14 topic categories by total questions across all years and exams.

Rank	Topic / Category	Total Qs	% of All	Priority Level
1	Networking & Data Communication	95	22.9%	★★★★★ Critical
2	Programming & Algorithms	56	13.5%	★★★★★ Critical
3	Hardware & Digital Logic	55	13.3%	★★★★★ Critical
4	Database & SQL	50	12.0%	★★★★★ Critical
5	Cybersecurity	38	9.2%	★★★★★ High
6	Software Engineering	24	5.8%	★★★ Medium
7	Data Structures	22	5.3%	★★★ Medium
8	Operating Systems	18	4.3%	★★★ Medium
9	Focus / Essay / Translation	15	3.6%	★★ Standard
10	Banking & Digital Finance	15	3.6%	★★ Standard
11	OOP Concepts	12	2.9%	★★ Standard
12	Cloud & Virtualization	10	2.4%	★★ Standard
13	Math & Number Systems	3	0.7%	★ Low
14	General Knowledge	2	0.5%	★ Low

Key Insight: Networking & Data Communication, Programming & Algorithms, Hardware & Digital Logic, and Database & SQL collectively represent **61.7%** of all exam questions. A candidate who masters these four areas is well-prepared for the majority of any paper.

3. YEAR-WISE QUESTION DISTRIBUTION

The matrix below shows how many questions from each topic category appeared per year. Use this to identify rising and declining trends.

Category	2020	2021	2022	2023	2024	2025	2026	Total
Networking & Data Communication	11	15	25	28	8	2	6	95
Programming & Algorithms	13	7	13	7	10	2	4	56
Hardware & Digital Logic	10	11	6	12	10	1	5	55
Database & SQL	8	10	12	6	10	2	2	50
Cybersecurity	2	4	6	12	4	2	8	38
Software Engineering	2	1	3	5	4	2	7	24
Data Structures	4	1	4	4	4	1	4	22
Operating Systems	1	8	2	1	2	2	2	18
Focus / Essay / Translation	5	2	0	0	0	0	8	15
Banking & Digital Finance	2	2	0	7	4	0	0	15
OOP Concepts	1	2	1	3	3	1	1	12
Cloud & Virtualization	0	0	0	5	3	0	2	10
Math & Number Systems	1	0	0	0	2	0	0	3
General Knowledge	0	0	0	0	0	0	2	2
YEAR TOTAL	60	63	72	90	64	15	51	415

3.1 Notable Trends Over the Years

- 2020–2021: Hardware & Digital Logic and Programming dominated, with heavy focus on circuit design, K-Maps, and coding problems.
- 2022: Networking questions surged to 25 — the highest single-year total — driven by multiple network engineering exam sets.
- 2023: Cybersecurity questions nearly tripled compared to 2021, reflecting industry concerns about digital banking security.
- 2023: Cloud & Virtualization appeared for the first time as a significant category (5 questions), linked to Bangladesh Bank's Maintenance Engineer exam.
- 2024: Programming, Database, and Hardware all tied at 10 each — exams became more balanced across technical domains.
- 2025: A smaller exam set (15 questions total) but covered all core categories evenly — likely a screening-level paper.
- 2026: Focus/Essay (8) and Cybersecurity (8) jointly topped the year, and Software Engineering doubled from prior years, suggesting a shift toward applied/written competency testing.

4. DETAILED TOPIC-BY-TOPIC ANALYSIS

This section provides a detailed breakdown of each topic category — including total frequency, percentage share, and the most commonly tested sub-topics.

4.1 Networking & Data Communication — 95 questions (22.9%)

Consistently the #1 topic across every year and every post type. Expect 3–6 networking questions in any Bank IT written exam.

Frequently Tested Sub-Topics
1. TCP/IP & OSI Model (layers, functions, protocols)
2. IP Addressing & Subnetting (CIDR, classful/classless)
3. Routing Algorithms (Dijkstra, distance vector, link-state)
4. HTTP, DNS, SMTP, SNMP, FTP protocols
5. TCP features (3-way handshake, congestion/flow control, ARQ)
6. LAN transmission media (twisted pair, fiber, coax)
7. Multiplexing (TDM, FDM, WDM, Synchronous/Statistical)
8. Network topologies (bus, ring, star, tree, mesh)
9. VPN, NAT, VLAN, DHCP
10. Bandwidth calculations, PCM/digitization problems
11. LTE/5G network elements, RFID, GGSN
12. ARP, CRC, Parity, Checksum error detection
13. Web caching, CDN
14. Hub vs Switch vs Router differences
15. IPv4 vs IPv6, public vs private addressing

High-frequency patterns: Subnetting/CIDR problems (appear in almost every exam), TCP mechanisms, OSI/TCP-IP model layers, and multiplexing calculations are nearly guaranteed questions.

4.2 Programming & Algorithms — 56 questions (13.5%)

The #2 category and the most diverse — questions range from simple coding to complexity analysis and graph algorithms.

Frequently Tested Sub-Topics
1. Sorting algorithms (Insertion, Bubble, Merge, Quick Sort)
2. Searching algorithms (Linear, Binary Search)
3. Recursion and Factorial/Fibonacci
4. Time & Space complexity analysis (Big-O)
5. Algorithm paradigms (Greedy, Divide & Conquer, Dynamic Programming)
6. C/C++/Java/Python coding problems
7. String manipulation (palindrome, reverse)
8. Array operations (sum, merge, prime finding)
9. Graph algorithms (Kruskal, Prim, Bellman-Ford, Dijkstra)
10. Output tracing of given code snippets
11. Catalan number, LCM, Armstrong number
12. Recursive merge of sorted arrays
13. Exception handling in Java
14. IP class identifier programs
15. Matrix operations

High-frequency patterns: Algorithm complexity (Big-O), sorting/searching implementations, and recursive programs appear in almost every exam set regardless of post.

4.3 Hardware & Digital Logic — 55 questions (13.3%)

#3 category — hardware questions are dominant in Engineer posts (ANE, AHE, AME) but also appear in Officer and SO IT exams.

Frequently Tested Sub-Topics
1. Boolean algebra & K-Map simplification
2. Logic gates (AND, OR, NOT, NAND as universal gate)
3. Combinational circuits (MUX, DEMUX, Half/Full Adder)
4. 7-Segment display decoder design
5. Cache memory organization (direct-mapped, tag/index/offset)
6. RAID levels (0, 1, 5, 10) and disk pack capacity calculations
7. SRAM vs DRAM, ROM types
8. SSD block/page structure
9. Microprocessor vs Microcontroller (8086, RISC vs CISC)
10. DAC/ADC conversion
11. Server hardware components and selection
12. Impact vs Non-Impact Printers
13. Transistor operation regions, JFET
14. Pipelining stages
15. Ergonomics and memory performance parameters

High-frequency patterns: K-Map simplification, logic circuit design, RAID levels, cache memory organization, and disk capacity calculations appear repeatedly.

4.4 Database & SQL — 50 questions (12.0%)

#4 category — heavily tested in ADA and Officer IT exams. SQL query writing is nearly universal.

Frequently Tested Sub-Topics
1. ER Diagram design (entities, relationships, cardinalities)
2. SQL queries (SELECT, JOIN, GROUP BY, HAVING, ORDER BY)
3. ACID properties of transactions
4. Normalization (1NF, 2NF, 3NF, BCNF)
5. DDL, DML, DCL commands
6. Transaction management and deadlock prevention
7. SQL vs NoSQL comparison
8. Triggers, stored procedures
9. Primary key, foreign key, unique key, indexing
10. Database failure classification
11. Distributed database systems
12. DBMS definition and purpose
13. Relational schema from ER diagram
14. Subqueries and aggregate functions
15. Database backup and recovery

High-frequency patterns: ER diagram design (soccer club / banking system are common scenarios), complex SQL with GROUP BY + HAVING, and ACID properties appear in most exam sets.

4.5 Cybersecurity — 38 questions (9.2%)

Rising trend: Only 2 questions in 2020, growing to 12 in 2023 and 8 in 2026. Expected to be a major focus area going forward.

Frequently Tested Sub-Topics
1. Cybersecurity CIA triad (Confidentiality, Integrity, Availability)
2. Cryptography (Symmetric vs Asymmetric, RSA, DH key exchange)
3. Firewall types (packet-filter, stateful, NGFW)
4. SQL Injection attacks and prevention
5. DDoS, Session Hijacking, XSS, CSRF attacks
6. Malware types (Ransomware, Trojan, Worm, Virus)
7. Digital Signatures and Certificates
8. SSL/TLS and Man-in-the-Middle attacks
9. Phishing and Social Engineering
10. Packet sniffing vs Packet spoofing
11. DMZ (Demilitarized Zone) architecture
12. Authentication vs Authorization
13. Vulnerability assessment and penetration testing
14. Access control (Blacklist/Whitelist, Explicit Deny)
15. Hashing vs Encryption differences

High-frequency patterns: CIA triad, SQL Injection with prevention, firewall types, and cryptography (symmetric vs asymmetric) appear across almost all recent exams.

4.6 Software Engineering — 24 questions (5.8%)

Mostly theoretical — SDLC models, testing levels, and increasingly AI/ML concepts in 2025–2026 exams.

Frequently Tested Sub-Topics
1. SDLC models (Waterfall, Spiral, Agile, Incremental)
2. Software testing levels (Unit, Integration, System, UAT)
3. Agile vs Waterfall comparison
4. Function point analysis and effort estimation
5. Debugging approaches
6. Software Quality Assurance (SQA)
7. DFA/NFA and Finite Automata construction
8. UML diagrams (Use Case, Class Diagram)
9. Machine Learning types (Supervised, Unsupervised, RL)
10. Generative AI and Explainable AI (XAI)
11. Version control (Git — commit vs push)
12. Requirements validation (functional vs non-functional)
13. Verification vs Validation (CMMI Level 3)
14. Deep Learning vs Federated Learning
15. Project management approaches

4.7 Data Structures — 22 questions (5.3%)

Consistent presence in AP and ANE posts. Focus is on practical operations rather than theoretical proofs.

Frequently Tested Sub-Topics
1. Binary Search Tree (BST) construction and traversal
2. Heap operations (Max-Heap Insert, Extract-Root)
3. Stack vs Queue (LIFO vs FIFO, applications)
4. Graph representations (adjacency matrix vs list)
5. Implementing Queue using two Stacks
6. Linked Lists (singly vs doubly)
7. Binary tree construction from In/Pre/Post-order
8. Hashing and collision resolution (linear probing)
9. Infix to Postfix conversion using Stack
10. Bubble Sort vs Merge Sort analysis
11. Graph isomorphism checking
12. Binary Search algorithm and requirements
13. Flowchart for algorithmic logic
14. Adjacency list for complete binary tree
15. In/out-degree calculation in directed graph

4.8 Operating Systems — 18 questions (4.3%)

Common in ASA and AHE posts. CPU scheduling with Gantt chart calculation is the most repeated question type.

Frequently Tested Sub-Topics
1. CPU Scheduling algorithms (FCFS, SJF, Round Robin, Priority)
2. Deadlock: conditions, prevention, detection
3. Memory management (Paging, Segmentation, Virtual Memory)
4. Page replacement algorithms (LRU, NRU, FIFO)
5. Process vs Thread (advantages of multithreading)
6. Producer-Consumer synchronization problem
7. Thrashing in demand-paged systems
8. Logical vs Physical address space calculation
9. Multiprogramming and context switching
10. Shell scripting (Linux/UNIX commands)
11. Memory partitioning (First-Fit vs Best-Fit)
12. Kernel objectives and types
13. Swapping and virtual memory interaction
14. Process scheduling Gantt charts
15. Rate Monotonic Schedulability analysis

4.9 Banking & Digital Finance — 15 questions (3.6%)

Present in every post type. Expected at SO IT and Officer IT level. More conceptual/essay-type than technical.

Frequently Tested Sub-Topics
1. Digital banking vs Traditional banking differences
2. IT role in the banking sector
3. Blockchain: working principle, benefits, use cases
4. E-commerce types (B2B, B2C, C2C)
5. Banking software used in Bangladesh
6. Digital finance and financial inclusion
7. IT Disaster Recovery Plan (DRP) and BCP
8. IS Audit and control policies
9. Web application deployment architecture
10. ICT innovations for online banking
11. System Administrator roles and responsibilities

4.10 Focus / Essay / Translation — 15 questions (3.6%)

Exclusively in BSCS-organized exams for Sonali Bank, Janata Bank, and assistant programmer posts. Bangla and English essays on IT/Banking topics are regular.

Frequently Tested Sub-Topics
1. Essays on COVID-19 and technology's role
2. Climate change and Bangladesh impact
3. Bangladesh graduation from LDC to developing country
4. Post-corona green recovery plans
5. Bangla/English translation passages
6. Role of IT in banking sector
7. Social media and rumors
8. AI application in banking sector (Bangla)
9. Data privacy and regulatory compliance (English)
10. NEET youth — economic/social impact
11. দুর্নীতিমুক্ত দেশ গঠনে যুবসমাজের ভূমিকা
12. Online Banking essay

4.11 OOP Concepts — 12 questions (2.9%)

Relatively consistent across years. Class diagrams, polymorphism, and inheritance are the most common sub-topics.

Frequently Tested Sub-Topics
1. OOP concepts: Encapsulation, Inheritance, Polymorphism, Abstraction
2. Types of inheritance (single, multiple, hierarchical, hybrid)
3. Method overloading vs Method overriding
4. OOP vs Procedural programming differences
5. Class diagrams (UML)
6. OOP bank account implementation
7. Circle class design (data + methods)

4.12 Cloud & Virtualization — 10 questions (2.4%)

Emerging category: First appeared significantly in 2023, and featured in 2024 and 2026 exams. Expected to grow in future exam cycles, especially for Senior Engineer posts.

Frequently Tested Sub-Topics
1. IaaS, PaaS, SaaS cloud service models
2. Virtualization and Hypervisor types (Type 1 vs Type 2)
3. Virtual Machine vs Container differences
4. Data center energy efficiency challenges
5. Cloud transformation strategy for banks
6. SaaS and multi-tenant architecture
7. Single vs multi-database for e-commerce

4.13 Math & Number Systems — 3 questions (0.7%)

Frequently Tested Sub-Topics
1. Binary to Decimal and Decimal to Hexadecimal conversions
2. Set operations (union, intersection, difference)
3. Probability with dice problems

4.14 General Knowledge — 2 questions (0.5%)

Only appears in BSCS-organized exams as a short GK section. Covers Bangladesh-specific IT and banking GK.

Frequently Tested Sub-Topics
1. Bangladesh banking history (first Bengali-owned bank)
2. International organizations (WIPO, CIRDAP)
3. Scientific inventions (dynamite, Nobel prizes)
4. Sports results and IT GK
5. Rohingya crisis management and international cooperation

5. QUESTION PATTERN BY POST / DESIGNATION

Different posts have distinctly different exam patterns. Use this section to tailor your preparation to your target post.

Post / Designation	Approx. Questions	Dominant Topics
Officer IT	76	Networking, Programming, Database, Cybersecurity, OOP
Assistant Programmer	72	Programming, Data Structures, OOP, Database, Focus/Essay
Senior Officer IT	40	Networking, Database, Hardware, Cybersecurity, Software Eng
Assistant Network Engineer	36	Networking, Database, Data Structures, Security, Hardware
ADA (Asst. Database Admin)	35	Database, Programming, Hardware, Security, Data Structures
Assistant Hardware/Maintenance Engineer	33	Hardware, Networking, Cloud, Security, Banking
Assistant System Administrator	14	OS, Networking, Security, Database, Programming
ANA (Asst. Network Admin)	13	Networking, Security, Database, Hardware, Programming
SO IT / ANSE / ANA / ANSE	61	Networking, Hardware, Security, Banking, Cloud
Other / Sub-Asst Engineer	45	Networking, Programming, Security, Hardware, OS

5.1 Officer IT / Senior Officer IT

- Full range of technical topics — no single area is overwhelmingly dominant.
- Networking (TCP/IP, subnetting, HTTP) and Database (SQL, ER diagrams) are always present.
- Cybersecurity questions increased sharply from 2022 onwards.
- Expect 1–2 programming coding questions alongside theory questions.

5.2 Assistant Programmer (AP)

- Programming and Data Structures are the core focus — expect 4–6 coding/algorithm questions.
- OOP concepts are tested more heavily here than any other post.
- BSCS-organized AP exams include Bangla/English essay and translation sections.
- Database SQL and security are secondary but consistent topics.

5.3 Assistant Network Engineer (ANE)

- Networking dominates — subnetting, routing algorithms, and protocol analysis are core.
- Database ER diagrams and SQL queries frequently appear (3–4 per exam).
- Hardware (logic circuits, combinational design) also appears regularly.
- One programming question (usually in C/C++) is standard.

5.4 Assistant Database Administrator (ADA)

- Database is the primary focus: ER diagrams, normalization, ACID, SQL queries, and triggers.
- Hardware (RAID levels, storage) and Programming are secondary topics.
- Security (data encryption, secure connections) features in recent exams.

5.5 Assistant Hardware / Maintenance Engineer (AHE/AME)

- Hardware (digital circuits, K-Maps, RAID, cache, ICs) is the dominant category.
- Networking (TCP/IP, multiplexing) and Cloud Computing are secondary.
- Banking IT topics (DR plans, audit policies, ICT applications) are important.

6. EXAM PREPARATION PRIORITY GUIDE

Based on frequency and trend analysis, here is a structured preparation roadmap:

Tier 1 — MUST MASTER (These 4 topics = 62% of all questions)

Topic	Key Areas to Cover
Networking & Data Comm (22.9%)	Subnetting/CIDR, OSI/TCP-IP layers, TCP mechanisms, routing algorithms, HTTP/DNS, multiplexing calculations
Programming & Algorithms (13.5%)	Big-O complexity, sorting (Merge/Quick), recursion, graph algorithms (Dijkstra/Kruskal), coding in C/Java/Python
Hardware & Digital Logic (13.3%)	K-Map simplification, logic circuits (MUX/DEMUX/Adders), RAID levels, cache organization, disk calculations
Database & SQL (12.0%)	ER diagrams, complex SQL (GROUP BY, HAVING, subqueries, JOINS), ACID properties, normalization (BCNF)

Tier 2 — HIGH PRIORITY (Rising trends + consistent presence)

Topic	Key Areas to Cover
Cybersecurity (9.2%) ↑	CIA triad, SQL injection, firewall types (NGFW), cryptography (RSA, DH), phishing, ransomware
Software Engineering (5.8%) ↑	SDLC models, Agile vs Waterfall, testing levels, AI/ML types (2025–26 trend), SQA
Data Structures (5.3%)	BST operations, Heap operations, Stack/Queue, linked lists, graph representations, hashing
Operating Systems (4.3%)	CPU scheduling with Gantt charts (FCFS, SJF, RR), deadlock conditions, paging/virtual memory, page replacement

Tier 3 — STANDARD (Post-specific importance)

Topic	Notes
Banking & Digital Finance (3.6%)	Important for Officer IT & SO IT. Focus on blockchain, digital banking, DRP/BCP, fintech trends.
Focus/Essay/Translation (3.6%)	Only in BSCS exams. Practice Bangla/English essays on IT, banking, and current affairs topics.
OOP Concepts (2.9%)	More important for AP posts. Inheritance types, polymorphism, encapsulation, class diagrams.
Cloud & Virtualization (2.4%) ↑	Growing trend. IaaS/PaaS/SaaS, Hypervisor (Type 1 vs 2), VM vs Container, cloud migration strategies.

7. EMERGING & FUTURE EXAM TRENDS (2025–2026 SIGNALS)

Based on the most recent 2025–2026 exam sets, the following topics show increasing prominence:

- Artificial Intelligence / Machine Learning (RL, DL, Federated Learning, Generative AI, XAI) — appeared in 2025 exams and is expected to grow significantly.
- Cloud Computing & Virtualization — now a standard topic for Engineer-level posts, expanding to Officer IT exams.
- API Standards (SOAP vs REST, Web Services) — appeared for the first time in 2025.
- Cybersecurity is now a Tier-1 topic — firewall rules, access control, ransomware response, and phishing are all being tested practically, not just theoretically.
- Software Engineering expanded to include AI ethics (Explainable AI), reflecting Bangladesh Bank's digital transformation agenda.
- Agile methodology and Version Control (Git) appeared in 2026 BSCS exams — likely to become standard for AP posts.
- RAID and server infrastructure knowledge is being tested more practically (real bank scenarios, not just theory).

8. QUICK REFERENCE — MOST REPEATED QUESTION TYPES

The following question types appear in 3 or more separate exam years:

Question Type	Years Found	Post Type
IP Subnetting / CIDR problems (create subnets, find hosts, broadcast)	2020,21,22,23,24,26	All posts
OSI / TCP-IP model layers and protocols	2020,21,22,23,26	All posts
SQL queries with JOIN, GROUP BY, HAVING	2020,21,22,23,24,25,26	All posts
ER Diagram design (banking/sports scenario)	2020,21,22,23,24,26	All posts
ACID properties of database transactions	2020,22,23,24,25	All posts
K-Map simplification and logic circuit	2020,21,22,23,24,26	ANE, AHE, AME, SO
CPU scheduling with Gantt chart (FCFS/SJF/RR)	2021,22,24,25	ASA, AHE, AME
Algorithm complexity analysis (Big-O)	2020,21,22,23,24,25	All posts
Sorting algorithm implementation	2020,21,22,23,24,25	Officer, AP
Disk pack / RAID capacity calculations	2021,22,23,24,26	AHE, SO IT
Cybersecurity CIA triad	2022,23,24,25,26	All posts
Firewall types and placement	2021,22,23,24,26	ANE, ANA, Officer
SQL Injection attack and prevention	2020,21,22,23,26	All posts
Symmetric vs Asymmetric encryption	2021,22,23,24	All posts
Binary Search Tree (BST) operations	2020,21,22,23	ANE, AP
Cryptography / RSA algorithm	2020,23,24,25	All posts
Deadlock conditions and prevention	2020,21,24	ASA, OS-focused
SDLC models (Waterfall vs Agile)	2022,23,24,25,26	Officer, AP
Digital banking vs traditional banking	2021,23,24,26	Officer, SO
IaaS / PaaS / SaaS cloud models	2023,24,26	AME, Senior posts